E-Governance Note

Name :
Roll No:
Semester:
Contact :
College :

Index

Unit 1

Concept of E-governance

Government:

A system or group of peoples responsible for making laws, maintaining order, and managing the affairs of country or state is called government. It is the authority that rules.

Governance:

The process of making decisions, implementing policies, and managing public affairs in an effective, transparent, and accountable manner. It defines how power and authority are exercised to achieve the welfare and development of society.

E-governance:

E-Governance refers to the use of Information and Communication Technology (ICT), especially the internet and digital systems, by the government to provide services, share information, and interact with citizens, businesses, and other government agencies efficiently.

It aims to improve transparency, accountability, efficiency, and public participation in governance through electronic means.

The importance / Advantages of E-governance are:

- Improves efficiency and speed of government services
- Increases transparency and accountability in administration
- Reduces corruption and paperwork
- Provides easy access to government services for citizens
- Saves time and cost for both government and public
- Encourages citizen participation in governance
- Improves communication between government and citizens
- Supports better decision-making through digital data management
- Enhances coordination between government departments
- Promotes good governance and digital development

The disadvantages of E-governance are:

- Requires high cost for setup and maintenance of ICT systems
- Depends heavily on internet and technology infrastructure
- Digital divide excludes people without internet or digital skills
- Cybersecurity risks like hacking and data theft
- Requires trained staff and continuous skill development
- System failures can disrupt government services
- Privacy concerns over citizen data usage
- Difficult implementation in rural and underdeveloped areas
- High initial investment and long implementation time
- Resistance to change from traditional government systems

Types of E-Governance:

The types of E-governance are:

1. Government to Government (G2G)
2. Government to Citizens (G2C)
3. Government to Business (G2B)
4. Government to Employee (G2E)

Government to Government (G2G):

Government to Government (G2G) e-governance refers to the electronic exchange of information, data, and services between different government departments, ministries, or agencies. It improves coordination, communication, and efficiency within the government system by using ICT tools and digital platforms.

In this model, different government bodies share data and work together through computerized systems, reducing duplication of work and improving decision-making and policy implementation.

Ex:

- Sharing data between Ministry of Health and Ministry of Finance for budget planning
- Communication between local government and central government for reporting and planning
- Police department sharing criminal data with border security
- Election commission coordinating with national ID database for voter verification

Government to Citizens (G2C):

Government to Citizen (G2C) e-governance refers to the electronic delivery of government services, information, and communication directly to citizens using ICT tools and digital platforms. It aims to make public services more accessible, efficient, and transparent.

In this model, citizens can interact with government systems online to access services, submit applications, and receive information without visiting government offices.

Ex:

- Online citizenship, passport, and license application systems
- Online tax payment services
- Online land record and property information systems
- Social security and welfare benefit applications online

Government to Business (G2B):

Government to Business (G2B) e-governance refers to the electronic exchange of information and services between government agencies and business organizations using ICT tools and digital platforms. It helps simplify business-related processes, improve transparency, and reduce delays in government approvals and regulations.

In this model, businesses can interact with the government online for registrations, licensing, tax payments, and other regulatory requirements.

Ex:

- Online business registration and licensing systems
- Online tax filing and payment (VAT, income tax, etc.)
- Customs clearance systems for import and export
- Online tendering and procurement systems

Government to Employee (G2E):
Government to Employee (G2E) e-governance refers to the use of ICT tools and digital systems to provide services and information between the government and its employees. It helps improve internal communication, manage employee records, and make administrative work more efficient and transparent.

In this model, government employees can access services like salary details, leave applications, training, and performance records through online systems.

Ex:
- Online Attendance system
- Online salary management systems
- Leave application and approval systems
- Online training and capacity-building programs
- Performance evaluation and reporting systems

Evolution of E-governance:
The evolution of E-governance has been a journey marked by various milestones and contributions from different regions and individuals. The E-governance evolution is given as:

1. **1970s**
   The concept of E-governance was first introduced with the use of computers in the government offices for data storage and record keeping.
2. **1980s**
   Government began using Information Systems and networking technologies to improve administrative work.
3. **1990s**
   With the growth of World Wide Web (WWW) and the internet, E-governance developed rapidly. Government started providing online information and services.
4. **2000s**
   Rapid development in ICT (Information and Communication Technology) increased online government services such as e-banking, e-tax, e-procurement, and online applications.
5. **Present Era**
   Government now uses advanced technologies like cloud computing, mobile apps, AI and Digital platforms to provide smart and transparent governance.

Scope of E-governance:
The scope of E-governance is quite broad and cover various aspects of government functions. They are as follows:

- **Service Delivery:**
  E-governance is used to provide government services to citizens in a faster, easier, and more efficient way through online platforms such as tax payment, license applications, and certificates.
- **Information Dissemination (Distribution):**
  It helps the government share important information like policies, notices, and public updates quickly and widely through websites, portals, and digital media.

- **Citizen Participation:**
  E-governance encourages citizens to take part in decision-making through online feedback, surveys, consultations, and grievance systems.
- **Internal Government Process Digitalization:**
  It improves internal operations by converting manual government processes into digital systems, making administration more efficient and less time-consuming.
- **Data Management:**
  It involves collecting, storing, and analyzing large amounts of government data in digital form to support planning, monitoring, and decision-making.
- **Security and Privacy:**
  E-governance ensures protection of sensitive government and citizen data through secure systems, encryption, and proper privacy policies.

Contents of E-governance:

The contents of E-governance include variety of digital elements. They are:

- **Government websites:**
  These are official online platforms that serves as the primary source of information about government services, policies, notices, and contact details.
- **Online forms:**
  These are digital application forms used by citizens to apply for services like passports, licenses, and certificates without visiting offices.
- **Digital platforms:**
  These include social media, mobile apps, and other online tools used for communication, awareness, and interaction between government and citizens.
- **E-services:**
  These are complete online government services such as tax payment, document issuance, registration, and other public services.
- **Data portals:**
  These are online systems that provide access to government data, statistics, reports, and open information for transparency and research.
- **Electronic ID systems:**
  These are digital identity systems used to uniquely identify citizens and securely access government services online.

Present Global Trends of Growth in E-governance:

Some present global trends of growth in E-governance are as follows:

- **Digital transformation and modernization:**
  Governments worldwide are shifting from traditional manual systems to fully digital systems. This improves efficiency, transparency, and speed of public service delivery.
- **Citizen-centric services:**
  E-governance is increasingly focused on citizens' needs, making services simple, accessible, and user-friendly. Governments design services based on convenience and feedback of citizens.

- **Mobile government (m-Government):**
  Many services are now available through mobile phones and apps, allowing citizens to access government services anytime and anywhere.
- **Emphasis on cybersecurity:**
  With increasing digital services, governments are focusing strongly on protecting data and systems from cyber threats, hacking, and misuse of information.
- **Cloud computing adoption:**
  Governments are using cloud technology to store, manage, and process large amounts of data efficiently, reducing cost and improving scalability.
- **International cooperation:**
  Countries are collaborating globally to share technology, best practices, and policies for improving e-governance systems.
- **Blockchain in governance:**
  Blockchain technology is being used to improve transparency, security, and trust in areas like land records, identity management, and financial transactions.

Evolution of E-governance in Nepal:

The development of e-governance in Nepal has been gradual, starting from basic computerization to more advanced digital services and online platforms.

- **Early initiatives (late 1990s–2000s):**
  The journey of e-governance in Nepal began with small-scale projects focused on computerizing and automating basic government processes in some departments.
- **Formation of HLCIT (2004):**
  The High-Level Commission for Information Technology (HLCIT) was established to promote ICT development and guide e-governance initiatives in Nepal.
- **National Information Technology Center (NITC):**
  NITC was formed as the main government IT agency to develop, manage, and implement e-governance systems and digital infrastructure.
- **Nepal Government Portal:**
  The official portal (www.nepal.gov.np) was launched to provide a single platform for accessing various government information and services.
- **Electronic Transaction Act (2063 B.S.):**
  This law provided a legal framework for electronic transactions, digital signatures, and cyber law in Nepal, supporting e-governance development.
- **Local e-governance initiatives:**
  Local governments started using digital systems for record keeping, service delivery, and communication with citizens.
- **Digital Nepal Framework (2018 AD):**
  This framework was introduced to promote digital transformation across sectors like education, health, agriculture, and governance.
- **Online services expansion:**
  Government services such as tax payment, registration, certificates, and applications are increasingly being provided through online platforms, making services more accessible and efficient. Government has also launched ‘Nagrik’ app to give services in a single online platform.

Difference Between:

E-Government	E-Governance
➢ E-Government refers to the use of ICT by government to provide services and information to citizens electronically.	➢ E-Governance is a broader concept that uses ICT to improve the overall governance system, including interaction between government, citizens, businesses, and organizations.
➢ It mainly focuses on delivering government services online.	➢ It focuses on improving the entire governance process and decision-making system.
➢ It is service-oriented.	➢ It is process-oriented and participation-oriented.
➢ It is mainly about government-to-citizen (G2C) services.	➢ It includes G2C, G2B (government to business), and G2G (government to government) interactions.
➢ It helps in providing faster and easier access to government services.	➢ It helps in improving transparency, accountability, and citizen participation in governance.
➢ Example: Online passport application, tax payment, and license renewal systems.	➢ Example: Online policy consultation, digital voting systems, and public feedback platforms.
➢ It is a subset of e-governance.	➢ It is a wider concept that includes e-government as one of its parts.
➢ It mainly improves efficiency in service delivery.	➢ It improves both governance quality and service delivery.

***

Unit 2

E-Governance Models

Model:
A model is a representation of a system or object that shows how it works. It serves as a framework that others can replicate to achieve similar results.

Concept of E-Governance Models:
E-Governance models are frameworks or approaches used by government to manage digital technologies, data, and information systems. They evolve continuously as new applications of ICT (Information and Communication Technology). Developing countries often experiment with different models to identify which best meet their governance needs. The purpose of these models is to enhance efficiency, transparency, and citizens participation in governance.

Types of E-governance / Digital Governance models:
1. Broadcasting Dissemination Model
2. Critical Flow Model
3. Comparative Analysis Model
4. Mobilization and Lobbying Model
5. Interactive Services Model / Government to Citizen to Government (G2C2G) Model

Broadcasting / Wider Dissemination Model: *[Dissemination –> Distribution]*
The broadcasting model is based on the mass dissemination of government related information that is already available in the public domain. Through ICT tools (like websites, portals, and media), the government spreads information widely to raise awareness among citizens about governance processes, public services, and how they can benefit from them. It emphasis transparency and accessibility, ensuring that citizens are informed about participations in governance.

The applications of Broadcasting Dissemination model are:
1. Publishing laws and rules online
2. Providing contact details of local, regional, and national government officials online.
3. Sharing government plans and reports online.
4. Disseminating (distributing) Legal decisions online

The merits of Broadcasting Model are:
1. Promotes transparency
2. Raises public awareness
3. Encourages accountability
4. Builds trust

The demerits of Broadcasting model are:
1. One way-communication as citizens receive information but cannot interact or give feedback.
2. Limited participation as citizens are not involved in decision making.
3. People without internet cannot access the information.

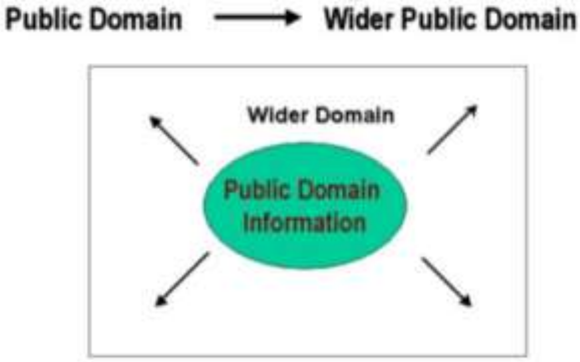


Fig: Broadcasting model

Critical Flow Model:

The critical flow model focuses on channeling information of critical value to a targeted audience or spreading it strategically in the wider public domain through ICT. It ensures that important information – such as corruption data, human rights reports, environment findings – reaches those who can act on it. The model removes barriers of distance and time, making information instantly accessible and impactful.

The applications of Critical Flow model are:

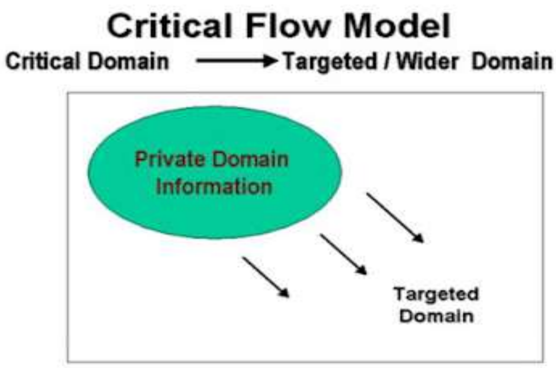
1. Publishing corruption related data about ministries or departments online for regulatory bodies.
2. Sharing research and inquiry reports commissioned by the government.
3. Providing human rights and criminal records to NGOs and concerned citizens.
4. Disseminating environmental information to local communities for awareness and action.

The merits of critical flow model are:

- Delivers important information to the right audience quickly
- Increases awareness about corruption, human rights, and public issues
- Promotes transparency and accountability in governance
- Helps expose weaknesses in government policies and decisions
- Encourages corrective actions and public participation

The demerits of critical flow model are:

- Government authorities may suppress or hide critical information
- Requires openness and transparency to work effectively
- Sensitive information may create political or social conflict
- Risk of misuse or spreading incomplete information
- Limited internet access may prevent some citizens from receiving information
- Dependence on ICT infrastructure and digital literacy



Comparative Analysis Model:

This model uses ICT and social media tools to compare governance performance across regions, institutions, or time periods. IT helps identify strengths and weaknesses in policies and governance practices, enabling improvement. IT has high potential for developing counties to learn from best practices and correct inefficiencies.

The applications of Comparative Analysis Model are:

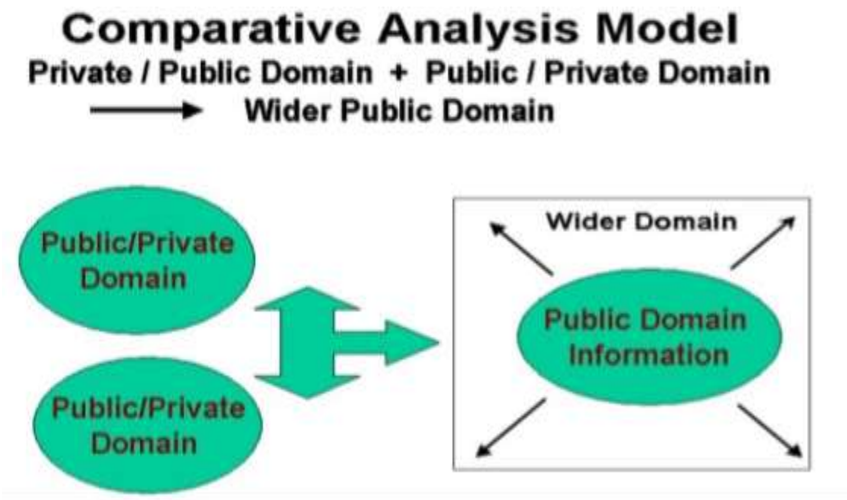
- ❖ Learning from past policies to improve future decision-making
- ❖ Evaluating policy effectiveness by comparing strengths and weaknesses
- ❖ Supporting informed, evidence-based decision making in governance
- ❖ Assessing performance of ministries, departments, or officials using data
- ❖ Monitoring trends in governance and policy changes over time

The merits of Comparative Analysis Model are:

1. Encourages transparency through measurable comparisons.
2. Helps in refining policies using past data.
3. Enables continuous monitoring for quality governance.

The demerits of Comparative Analysis Model are:

1. It requires analytical ability for analysis and strong argumentation.
2. Ineffective if public participation is weak.



Mobilization and Lobbying Model:

It is a two-way or multi-way communication model of e-governance in which governments, organizations, or interest groups collect opinions, suggestions, and feedback from citizens through digital platforms. The gathered information is analysed and used to modify policies, improve decisions, and address public needs.

For example, governments may use online forms, surveys, social media, or discussion portals to collect public opinion before finalizing a policy or plan. This model is widely used because it creates a planned and organized flow of information between the government and citizens, helping to build strong virtual partnerships and encourage public participation in governance.

The applications of Mobilization and Lobbying model are:

1. Online public awareness campaigns
2. Digital petitions and signature collections
3. Social media movements for public issues
4. Citizen participation in policy making
5. Online discussions and public opinion collection
6. Advocacy for human rights, environment, education, etc.

The merits of Mobilization and Lobbying model are:

1. Increases citizen participation in governance
2. Makes government more accountable
3. Helps spread information quickly
4. Encourages transparency and democracy
5. Easy and low-cost communication with large audiences
6. Gives voice to minority or ignored groups

The demerits of Mobilization and Lobbying model are:

1. False information can spread rapidly
2. Online campaigns may create social conflicts
3. Cyber bullying and misuse of social media
4. Digital divide may exclude rural or poor citizens
5. Sometimes used for political manipulation
6. Privacy and security issues may rise.

Iterative Service Model (G2C2G):

The Interactive Service Model is a two-way communication model of e-governance that establishes direct interaction between the government and citizens through digital technologies. In this model, citizens not only receive services from the government but also provide feedback, share opinions, and participate in decision-making processes.

It strengthens the relationship between government and citizens by creating interactive communication channels such as online portals, mobile applications, video conferencing, online discussions, and digital service platforms. For example, a government tax payment website allows citizens to pay taxes online, ask questions, submit complaints, and communicate directly with government agencies. This model promotes active public participation, transparency, and collaborative governance through continuous digital interaction.

The applications of Iterative Service Model are:

- ❖ Online tax payment and government service portals
- ❖ Electronic voting and online elections
- ❖ Public debates and policy discussions
- ❖ Citizen feedback and complaint systems
- ❖ Video conferencing between citizens and policymakers
- ❖ Online discussion forums and consultation platforms
- ❖ Decentralized governance through local digital participation

The merits of Iterative Service Model are:

- ❖ Provides direct and interactive government services
- ❖ Encourages citizen participation in governance

- ❖ Improves transparency and accountability
- ❖ Enables real-time communication and feedback
- ❖ Makes government services faster and more convenient
- ❖ Strengthens government–citizen relationships

The demerits of Iterative service Model are:

- ❖ Requires strong ICT infrastructure and internet access
- ❖ Implementation and maintenance can be costly
- ❖ Digital divide may exclude rural or poor citizens
- ❖ Security and privacy risks in online interactions
- ❖ Technical failures can interrupt services and communication

Maturity Model for E-governance:

A maturity model is a framework used to evaluate how developed and efficient an organization's digital systems and processes are. This model helps understand their current level of digital transformation. This model identifies gaps, highlights weaknesses, and provides direction for improving efficiency, transparency, and service delivery in government organizations.

The benefits/ Applications of Maturity Model in e-governance are:

1. Compares digital maturity of different government departments or ministries.
2. Provides clear direction for improving e-governance systems and processes.
3. Uses standard benchmarks for fair and objective evaluation
4. Helps in ranking e-governance portal based on their development level
5. Identifies gaps and weaknesses in existing systems
6. Supports better planning and decision making for digital transformation

The 5 maturity stages of E-governance are:

1. Level 1 – Closed
2. Level 2 – Initial
3. Level 3 – Planned
4. Level 4 – Realized
5. Level 5 – Optimized *(or, institutionalized)*

Level 1 – Closed:

The Closed stage is the most basic level of the Maturity Model in e-governance. At this stage, the organization does not use Information and Communication Technology (ICT) for governance activities and has no clear plan or vision for digital transformation. All processes are handled manually using traditional methods.

The characteristics of level 1 e-governance phase are:

- ❖ No use of ICT in governance or service delivery
- ❖ Lack of awareness or interest in e-governance
- ❖ No digital strategy or planning for future development
- ❖ Fully manual systems for records and administration
- ❖ No online services available for citizens
- ❖ Very low efficiency and slow service delivery
- ❖ Limited transparency and communication with the public

Level 2 – Initial:
The Initial stage is the early phase of e-governance maturity where an organization has just started adopting ICT in some areas. However, the use of technology is not well planned or coordinated. There is no clear strategy for digital transformation, and most efforts are temporary or project-based rather than systematic.

The characteristics of level 2 e-governance phase are:

- Partial use of ICT in some departments or activities
- Lack of overall strategy or long-term e-governance plan
- Uncoordinated and individual automation efforts
- Limited integration between different government departments
- Some basic online services may be available but not fully developed
- Low consistency and continuity in digital initiatives
- Dependence on manual systems is still very high

Level 3 – Planned:
The Planned stage is an advanced transitional phase in e-governance maturity where the organization begins to adopt a structured and systematic approach to ICT implementation. At this level, there is a clear vision, defined goals, and proper planning for digital transformation. Government departments start integrating electronic services in a more coordinated way, moving beyond isolated efforts toward a more unified system.

The characteristics of level 3 e-governance phase are:

- Clear vision and strategic planning for e-governance
- Systematic use of ICT across multiple departments
- Initial integration of electronic services between departments
- Improved coordination and data sharing among government units
- Development of structured digital governance policies
- Better organization of services compared to earlier levels
- Focus on expanding and improving online service delivery

Level 4 – Realized:
The Realized stage is a mature phase of e-governance where the organization successfully implements fully functional e-governance systems. At this level, ICT is widely deployed and government services are integrated across departments. The focus is on efficient, reliable, and citizen-friendly service delivery through connected digital systems. Most processes are automated, and different government units work in coordination through shared platforms.

The characteristics of level 4 e-governance are:

- Fully implemented e-governance systems across major government departments
- High level of integration between different services and databases
- Automated workflows for faster service delivery
- Improved efficiency, transparency, and accountability
- Reduced manual work and paperwork
- Citizens can access most services online in a connected system
- Strong coordination between departments through ICT platforms

Level 5 – Optimized:
The Optimized stage is the highest level of e-governance maturity where the organization achieves a fully advanced, intelligent, and continuously improving digital governance system. At this stage, ICT is

not only fully integrated but also used for real-time monitoring, decision-making, and service optimization. The system becomes highly citizen-centric, adaptive, and efficient, with continuous improvements based on data, feedback, and emerging technologies.

The characteristics of level 5 e-governance are:

- Fully integrated and highly advanced e-governance system
- Continuous monitoring and improvement of services
- Use of smart technologies (AI, data analytics, automation)
- Highly efficient, transparent, and responsive governance
- Strong focus on citizen satisfaction and participation
- Predictive and proactive decision-making
- Minimal manual intervention in government processes

ICT:

ICT stands for Information and Communication Technology. It refers to all digital technologies such as computers, internet, mobile phones, networks, software, etc. used to collect, store, process, and share information.

E-governance toward Good Governance or,
How we achieve good governance through E-governance?

E-governance plays a key role in achieving good governance by using ICT to make government processes more efficient, transparent, accountable, and citizen friendly. E-governance achieves good governance as:

- **Transparency:** Digital systems reduce corruptions by making information easily accessible.
- **Accountability:** Online records and tracking systems make government actions more traceable
- **Efficiency:** Services become faster and more accurate due to automation.
- **Responsive Governance:** Government becomes quicker in responding to citizens needs and issues.
- **Accessibility:** Government services become available anytime and anywhere.
- **Citizen participation:** People can give feedback and participate in decision making through online platforms.
- **Cost reduction:** Reduces paperwork and operation costs.
- **Better decision making:** Data driven system help in making informed policies.

***

Unit 3

E-Governance Infrastructure, Stage in Evolution and Strategic for Success

E-readiness:

E-readiness means how ready a country or organization is to use ICT (like computers, internet, and software) to improve its economy, services, and governance. It shows how well prepared they are to adopt digital systems for better development and public services.

In e-governance, it means how ready the government is to use digital systems for providing online services and managing information.

The areas that ensure the government is ready to adopt and manage digital system effectively are called e-readiness preparedness. The e-readiness preparedness types are:

1. Data system Infrastructure Preparedness
2. Legal Infrastructure Preparedness
3. Institutional Infrastructure Preparedness
4. Human Infrastructure Preparedness
5. Technical Infrastructure Preparedness

Data System Infrastructure Preparedness:

Data System Infrastructural Preparedness refers to the readiness of an organization to manage government data in electronic form for effective e-governance. The core of e-governance is the Electronic Management Information System (E-MIS), which stores, processes, and manages information digitally.

For successful e-governance, data that were previously managed manually must be computerized and converted into electronic form. This requires a well-prepared computerized database or data warehouse system. Data quality and data security are major concerns, especially in developing countries where government ICT infrastructure may not be fully developed.

This is one of the main computerization activities in any government process and may take several years to implement successfully.

The key components of Data system Infrastructure Preparedness are:

1. Converting paper-based records and processes into digital form.
2. Developing proper systems to store and manage electronic data efficiently.
3. Ensuring data is accurate, complete, consistent, and reliable.
4. Protecting government data from unauthorized access, misuse, or cyber threats.

Legal Infrastructure Preparedness:

Legal Infrastructure Preparedness refers to the readiness of a country's legal system, laws, policies, and institutions to support and manage e-governance effectively. It ensures that proper legal frameworks exist to handle challenges such as digital transactions, cybersecurity, privacy, emergencies, administrative reforms, and changing social needs.

For successful e-governance implementation, governments need updated laws and regulations that support the reengineering of existing business processes, administrative rules, and digital governance practices.

Many developing countries still lack sufficient legal infrastructure for smooth digital transformation, while developed countries have achieved greater success through strong administrative reforms and legal support for e-governance.

The key components of Legal Infrastructure Preparedness are:

- **Laws and regulations:** Having clear, updated laws to manage issues such as digital security, privacy, public safety, human rights, and electronic transactions.
- **Administrative and business process reforms:** Modifying traditional government procedures and regulations to support digital governance systems.
- **International cooperation:** Participating in international agreements and collaboration to handle global legal and cybersecurity issues.
- **Public awareness and education:** Educating citizens about legal rights, digital responsibilities, and the proper use of e-governance services.
- **Cyber laws and data protection:** Establishing legal measures to protect electronic data, privacy, and online communications.

Institutional Infrastructure Preparedness:

Institutional Infrastructure Preparedness refers to the ability of government organizations and institutions to effectively respond to challenges, crises, and emergencies while successfully implementing e-governance systems. It includes the overall readiness of institutions in terms of physical infrastructure, resources, policies, coordination systems, and trained personnel required for smooth digital governance.

For successful e-governance implementation, strong institutional support is essential. Government institutions must have proper infrastructure, sufficient resources, and well-trained staff to manage digital systems and ensure efficient service delivery.

The key components of Institutional Infrastructure Preparedness are:

- **Physical infrastructure:**
  Availability of necessary facilities such as ICT equipment, internet connectivity, data centers, and communication networks required to support e-governance systems.
- **Resources (financial, human, and material):**
  Adequate funding, skilled workforce, and required materials are essential for implementing and maintaining e-governance projects successfully.
- **Training and capacity building:**
  Regular training programs, workshops, drills, and simulations are needed to develop skilled personnel who can efficiently handle digital systems and respond to challenges.
- **Coordination mechanisms:**
  Strong coordination between different government departments and institutions is necessary to ensure smooth communication, data sharing, and effective service delivery.
- **Policies and institutional support:**
  Clear policies and supportive institutional frameworks help guide the implementation and sustainability of e-governance systems.

Human Infrastructure Preparedness:

Human Infrastructure Preparedness refers to the ability of individuals, organizations, and communities to effectively use their human resources to achieve goals and manage changes in governance systems. In the context of e-governance, it focuses on ensuring that people have the necessary skills, knowledge, and capabilities to actively participate in and support digital government processes.

It is essential because even with advanced technology, e-governance cannot succeed without trained and capable people who can operate systems, manage services, and contribute to decision-making.

The key components of Human Infrastructure Preparedness are:

- **Capacity building:**
  Providing training, education, and skill development programs to improve the abilities of government officials, civil servants, and citizens in using ICT and e-governance systems.
- **Leadership development:**
  Developing strong leadership skills within government institutions to guide digital transformation and ensure effective implementation of e-governance policies.
- **Citizen participation and engagement:**
  Encouraging active involvement of citizens in governance processes through digital platforms, feedback systems, and online participation tools.
- **Ethical and professional standards:**
  Promoting honesty, transparency, accountability, and professionalism among government employees and institutions.
- **Diversity and inclusion:**
  Ensuring that people from all backgrounds are included in decision-making processes and have equal access to e-governance services.

Technical or Technological Infrastructure Preparedness:

Technical Infrastructure Preparedness refers to the readiness of a country, organization, or institution to effectively use and support technological systems for governance, service delivery, communication, and economic development. It includes the availability, reliability, scalability, security, and interoperability of hardware, software, networks, and other digital resources required to implement and maintain technology-based e-governance systems.

In simple terms, it ensures that the technical systems and tools needed for e-governance are strong, stable, and capable of handling digital services efficiently.

The key components of Technological Infrastructure Preparedness are:

- **Digital infrastructure:**
  Availability of reliable internet connectivity, data centers, and cloud computing services to support storage, processing, and delivery of e-governance applications.
- **Interoperability:**
  Different government systems and databases should be able to communicate and exchange data smoothly to ensure integrated and efficient service delivery.
- **Cybersecurity:**
  Protection of government systems and sensitive data from cyber threats using tools such as encryption, firewalls, intrusion detection systems, and regular security audits.

- **Data privacy:**
  Ensuring strict policies and protocols to protect citizens' personal information from unauthorized access or misuse.
- **Scalability and flexibility:**
  The technical infrastructure should be able to handle increasing user demand and adapt easily to new technologies and changing requirements.

Why do we need e-readiness?

We need e-readiness to ensure a country or organization is prepared to use ICT effectively for e-governance and development. It helps improve service delivery, efficiency, transparency, and citizen participation. It also ensures proper infrastructure, skills, and systems are in place for successful digital transformation.

Evolutionary Stages of E-governance:

E-governance develops gradually from simple use of ICT tools to fully integrated digital governance systems. The level of development may vary from country to country depending on infrastructure, policies, and digital readiness. The various stages of E-governance are:

1. **Use of Email and Internal Networking**
   The most basic and widely used ICT tool in government is email. It is simple, cheap, and effective for communication within departments. Internal networking (intranets) also helps government offices share information easily. This stage mainly focuses on improving internal communication.
2. **Use of Internet for Government Activities**
   At this stage, government offices begin using the internet for official work, information access, and communication. Employees use web browsing for both professional and general purposes. Many government departments such as agriculture, finance, and education start establishing an online presence.
3. **Public Access to Information**
   Government departments create websites or web pages to provide information to citizens. People can access public data such as education details, tourism information, census data, and statistical reports. This improves transparency and awareness.
4. **Two-Way Interactive Communication**
   At this stage, citizens can interact with government departments through online systems. They can ask questions, submit forms, give feedback, and communicate with officials through digital platforms. It enables basic transactions and interaction between government and citizens.
5. **Online Transactions**
   Government services become more advanced, allowing citizens to perform online transactions such as payments for taxes, bills, licenses, and other services. This improves convenience and reduces manual processes.
6. **Enriched Digital Democracy**
   Citizens actively participate in governance through digital platforms such as online voting, surveys, consultations, and policy discussions. This strengthens democratic participation and decision-making.

7. **Integrated or Joined-Up Government**
   At the highest level, all branches of government – executive, legislature, and judiciary – are digitally connected. Systems are fully integrated, allowing seamless information sharing, coordination, and efficient governance across all institutions.

***

Unit 4

Application of Data Warehouse and Data Mining in Government

Data Warehouse:

A Data Warehouse is a large centralized system used to store a huge amount of data collected from different sources within an organization. It organizes data in a structured way so that it can be easily accessed, analysed, and used for decision-making.

In e-governance, a data warehouse stores data from various government departments (like health, education, finance, etc.) in one place. This helps the government to analyse information, generate reports, and make better policies based on accurate and complete data.

The features of Data warehouse are:

- **Subject-oriented:**
  Data is organized based on specific subjects like health, education, finance, or population, etc. instead of daily transactions. It helps users easily access and analyze data related to a specific subject for better decision making.
- **Integrated:**
  Data is collected from different sources (departments) and combined into a single, consistent format. This removes duplication and inconsistency.
- **Time-variant:**
  Data is stored with a time reference, meaning it keeps historical records. This helps in analyzing trends and changes over time.
- **Non-volatile:**
  Once data is stored in the data warehouse, it is not frequently changed or deleted. It is mainly used for reading, analysis, and reporting rather than daily updates.
- **Multidimensional View:**
  A data warehouse stores data in such a way that the data can be viewed at from different angles, called dimensions like time, place, product, etc. This is called a multidimensional view.
- **Supports OLAP and Data Mining:**
  Data warehouse supports OLAP (Online Analytical Processing) that let users explore and analyse data interactively. They also support data mining, which help in discovering hidden patterns, trends and relationship in the data.

The advantages of Data Warehouse are:

- Stores large amount of data in one centralized system
- Integrates data from multiple departments or sources
- Improves data consistency and reduces duplication
- Helps in better decision-making through easy data analysis
- Supports historical data analysis (trend analysis over time)
- Makes reporting faster and easier
- Improves efficiency in e-governance and business processes

The disadvantages of Data Warehouse are:
- Development and maintenance cost is high
- Designing and managing the system is very complex
- Skilled professionals are required for operation
- Time-consuming process to build and update
- Data may become outdated if not regularly updated
- Security risks if not properly protected

The components of data warehouse are:
- **Data Sources:** Systems from which data is collected, such as government departments, databases, files, and applications.
- **ETL Process (Extract, Transform, Load):** Extracts data from different sources, converts it into a standard format, and loads it into the data warehouse.
- **Data Warehouse Database:** The central storage where cleaned, structured, and integrated data is stored for analysis.
- **Metadata:** Data about data, which describes the structure, source, and meaning of stored information.
- **Data Marts:** Smaller, department-specific sections of the data warehouse (Ex: health, education, finance).
- **Query and Reporting Tools:** Tools used by users to retrieve data, perform analysis, and generate reports for decision-making.

Difference Between:

OLAP (Online Analytical Processing)	OLTP (Online Transaction Processing)
➢ Used for data analysis and decision-making	➢ Used for daily transaction processing
➢ Handles large volume of historical data	➢ Handles current and operational data
➢ Focuses on complex queries and reports	➢ Focuses on simple, fast transactions
➢ Used by managers and analysts	➢ Used by clerks, cashiers, users, etc.
➢ Data is less frequently updated	➢ Data is continuously updated in real-time
➢ Examples: sales analysis, trend reporting	➢ Examples: bank transactions, ticket booking
➢ Optimized for read operations	➢ Optimized for insert, update, delete operations

Data Mining:
The process of discovering hidden patterns, relationships, trends, and anomalies from large datasets using techniques like classification, clustering, regression, etc. is called data mining. It is the core step in the larger process of Knowledge Discovery in Database (KDD). The goal is not just to extract data, but to find hidden insights that can help in decision making, prediction, and problem solving.

The advantages of Data mining are:

- Helps in discovering useful patterns and trends from large data sets
- Improves decision-making process
- Increases business efficiency and performance
- Helps in customer behavior analysis and marketing strategies
- Useful in fraud detection (banking, insurance, etc.)
- Saves time by analyzing large data automatically
- Supports predictive analysis and forecasting

The disadvantages of Data mining are:

- Privacy issues I.e. risk of personal data being misused
- High cost of tools and implementation
- Skilled professionals are required to analyze data
- Possibility of inaccurate or misleading results
- Security risks if data is not properly protected
- Complex process for large and unorganized data
- Ethical issue i.e. misusing people's personal information without their consent. It may also lead to excessive monitoring individuals, which raises privacy and fairness issues.

National Data Warehouses:

A National Data Warehouse in Nepal is a centralized system that collects and stores large amounts of data from different government ministries, departments, and major organizations. It allows researchers, planners, and policy makers to access historical and integrated data from various sectors. In Nepal, organizations such as Singha Durbar Data Center, Agriculture Development Bank, Nepal Telecom, and Nepal Stock Exchange (NEPSE) already use data warehouse systems to manage and analyse large datasets for decision-making and service improvement.

The key characteristics and purpose of National Data warehouse are:

- **Data integration:** Combines data from different government ministries and organizations into one unified system.
- **Comprehensive data coverage:** Covers data from multiple sectors like health, education, agriculture, finance, and communication.
- **Decision support system:** Helps government officials and policy makers make informed and data-driven decisions.
- **Policy formulation and evaluation:** Supports creation of new policies and evaluation of existing government programs using real data.
- **Historical data storage:** Stores past data for trend analysis and long-term planning.
- **Data security and privacy:** Ensures sensitive government and citizen data is protected from unauthorized access.
- **Standardized data format:** Ensures all data from different sources is stored in a consistent format for easy use.
- **Improved data accessibility:** Allows authorized users to access data quickly from a single platform.
- **Better planning and forecasting:** Helps government predict future trends in population, economy, and resources.

- ❖ **Supports transparency:** Makes government data more organized and reliable for research and public use.
- ❖ **Efficiency in governance:** Reduces duplication of data and improves coordination between departments.

Area for Data Warehouse and Data Mining: *[Applications/ fields]*

The Application areas of Data warehouse and Data mining are given as:

- ❖ **Public Finance and Revenue Management**

  Data Warehouse:
  Used to store large financial data such as taxes, budgets, subsidies, and government transactions in a centralized system for reporting and analysis.

  Data Mining:
  Data mining is used to detect tax evasion by analyzing transaction patterns. It helps in budget forecasting by studying past revenue and expenditure trends. It also identifies fraud in subsidy and welfare programs through anomaly detection and supports auditing by combining financial data from different departments.

- ❖ **Public Health and Healthcare**

  Data Warehouse:
  Used to store patient records, hospital data, disease reports, and health statistics in one system for easy access and management.

  Data Mining:
  Data mining helps in early detection of disease outbreaks by analyzing health data patterns. It is used to predict hospital resource needs such as beds, staff, and medicines. It also tracks vaccination coverage and supports medical research through pattern analysis.

- ❖ **Law Enforcement and Public Safety**

  Data Warehouse:
  Used to store crime records, investigation data, criminal databases, and border security information in a centralized system.

  Data Mining:
  Data mining helps identify crime patterns and high-risk areas. It is used for fraud and corruption detection across agencies. It also detects cyber threats and flags suspicious travel or cargo activities for better security.

- ❖ **Education Management**

  Data Warehouse:
  Used to store student records, exam results, attendance, and institutional data for analysis and reporting.

  Data Mining:
  Data mining is used to predict student dropouts and identify at-risk students. It helps compare school performance, improve learning outcomes, and detect fraud in scholarships and grants.

- **Agriculture and Rural Development**

  Data Warehouse:
  Used to store agricultural data such as crops, land records, subsidies, and weather information for planning and analysis.

  Data Mining:
  Data mining helps predict crop yield using weather and soil data. It tracks subsidy distribution, detects leakage, and analyzes market price trends for better farmer support.

- **Infrastructure and Urban Planning**

  Data Warehouse:
  Used to store data related to transport, utilities, housing, and city infrastructure in a structured system.

  Data Mining:
  Data mining supports smart city planning by analyzing traffic and IoT data. It predicts infrastructure failures, studies urban growth, and analyzes resource usage like water and electricity.

- **Election and Citizen Services**

  Data Warehouse:
  Used to store voter records, census data, and citizen service information in a centralized system.

  Data Mining:
  Data mining helps clean voter lists, analyze citizen complaints, and improve public services. It also studies census and service usage data for better governance planning.

- **Judiciary and Legal Administration**

  Data Warehouse:
  Used to store court case records, legal documents, and judicial data for easy access and tracking.

  Data Mining:
  Data mining is used to analyze case backlogs, improve court efficiency, and identify legal patterns. It also supports legal research and ensures consistency in judgments across courts.

Big Data in E-Governance:

Big Data refers to extremely large and complex datasets that cannot be easily processed using traditional data processing tools or methods. These data sets are so huge and fast-growing that special technologies are required to store, manage, and analyze them. The process of examining such large data to discover hidden patterns, trends, and useful information is called Big Data Analytics.

In e-governance, Big Data helps governments improve services and decision-making by analyzing large amounts of data from citizens, businesses, social media, sensors, and public records. It supports better policy making, faster service delivery, and more effective response to public needs. It also integrates different government databases, improving coordination, transparency, and efficiency in governance.

Difference Between:

Data Warehouse	Data Mining
➢ A system used to store and manage large amounts of data	➢ A process used to analyze data and discover useful patterns
➢ Focuses on data storage and organization	➢ Focuses on data analysis and knowledge discovery
➢ Collects data from multiple sources into one centralized system	➢ Extracts meaningful information from stored data
➢ Used for reporting and decision support	➢ Used for prediction, trend analysis, and pattern detection
➢ Stores historical and integrated data	➢ Finds hidden relationships and trends in data
➢ Main purpose is efficient data management	➢ Main purpose is gaining useful insights from data
➢ Example: Government database storing citizen records	➢ Example: Detecting fraud from transaction patterns

***

Unit 5

E-Governance of Nepal

Government Enterprise Architecture (GEA):

- Government Enterprise Architecture (GEA) is a structured framework that provides guidelines, architectural principles, development methods, and reference models for government business and technology services. It helps different government departments work together efficiently and ensures better and faster service delivery to citizens and businesses.
- The Government of Nepal has adopted The Open Group Architecture Framework (TOGAF) as the standard framework for developing its Government Enterprise Architecture.
- TOGAF helps in developing architectures that are consistent, well-organized, and based on international best practices. It also ensures that the needs of stakeholders, current requirements, and future government goals are properly addressed.
- TOGAF provides a practical and standardized approach for implementing Enterprise Architecture (EA). It supports the development of open, flexible, and interoperable systems that improve organizational efficiency and service management.
- TOGAF 9.0 has been adopted as the foundation of the Nepal GEA framework. The framework is customized and tailored according to the specific requirements and needs of the Government of Nepal.
- Architecture Development Method (ADM) is the core component of TOGAF that provides a systematic and step-by-step approach for developing, implementing, and managing Enterprise Architecture. It consists of a series of interconnected phases that support the complete life cycle of architecture, covering activities from planning and design to implementation, operational deployment, maintenance, and change management.

The Nepal government Enterprise Architecture is given as: **[or, TOGAF of Nepal]**

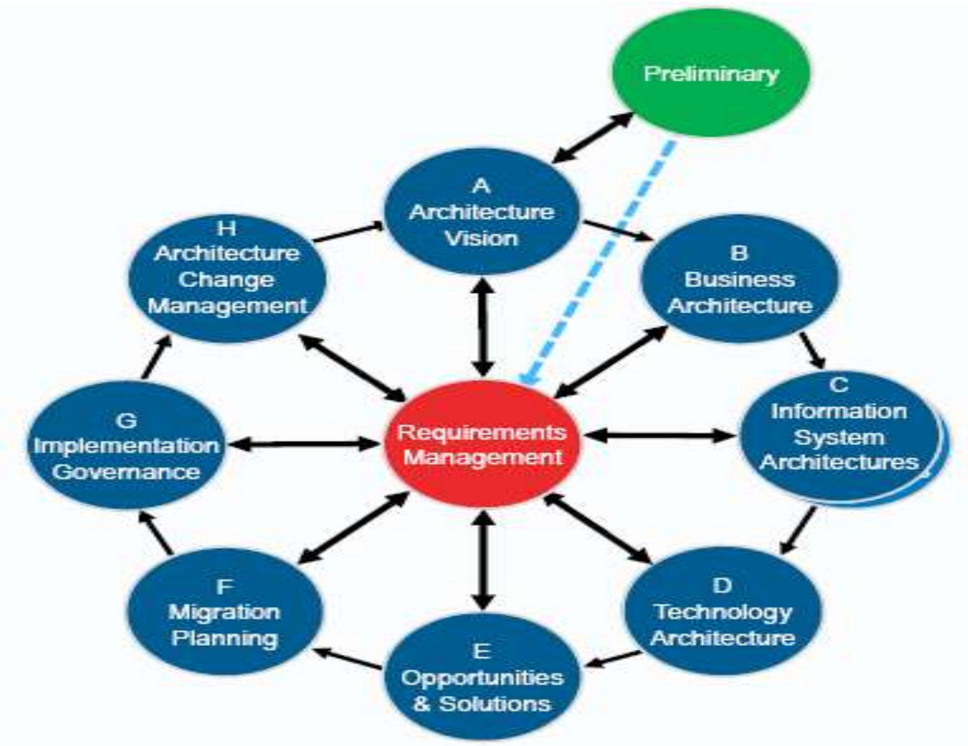


Fig: Nepal Government Enterprise Architecture

The phases of Nepal Government Enterprise Architecture are given as:

1. **Preliminary Phase:**
   The Preliminary Phase is the first step of TOGAF ADM that prepares the foundation for enterprise architecture by defining the framework, principles, governance, and roles. It sets up the environment for enterprise architecture development.
2. **Architecture Vision:**
   Defines the overall goal, scope, and vision of the architecture. It answers what we want to achieve.
3. **Business Architecture:**
   Describes the business processes, organizational structure, and how government services are delivered.
4. **Information Systems Architecture:**
   Focuses on data and application architecture, showing how information is stored, managed, and used.
5. **Technology Architecture:**
   Defines the hardware, software, networks, and technical infrastructure needed.
6. **Opportunities and Solutions:**
   Identifies possible solutions and projects to implement the designed architecture.
7. **Migration Planning:**
   Plans how to move from the current system to the target architecture in a phased manner.
8. **Implementation Governance:**
   Ensures that implementation follows defined architecture standards and guidelines.
9. **Architecture Change Management:**
   Manages and controls changes and updates to the architecture over time.

E-Governance Master Plan (EGMP):

- The EGMP is a blueprint for integrating Information and Communication Technology (ICT) in government to improve efficiency, transparency, and service delivery.
- It builds on past experiences and existing plans while setting a clear vision for future e-government development.
- It addresses not only technical issues but also organizational, legal, and policy challenges in implementing e-government.
- It includes key areas such as vision and objectives, public awareness and adoption, infrastructure development, service delivery improvement, cyber security, and policy and legal frameworks.
- It also focuses on improving coordination between government organizations and strengthening governance systems for effective implementation.
- The EGMP was jointly developed by the High-Level Commission for Information Technology (HLCIT) and the National Information Technology Centre (NITC) with technical assistance from the Korea IT Industry Promotion Agency (KIPA).

Data Centre:

A Data Centre is a centralized facility where computing and networking resources are stored and managed. It is used for collecting, storing, processing, and routing large volumes of data in an organized and secure manner.

A data centre consists of key components such as servers, routers, switches, firewalls, storage systems, and application delivery systems. It provides three main resources: network infrastructure, storage infrastructure, and computing resources.

Data centre services are designed to ensure high performance, reliability, and data security. Security tools like firewalls and intrusion detection systems are used to protect data and prevent unauthorized access.

Government Integrated Data Centre (GIDC):

The Government Integrated Data Centre (GIDC / GTDC) is the main government-owned data centre in Nepal. It is the central hosting facility that provides infrastructure for government websites, portals, applications, and other IT services.

It was established in 2008 AD with the collaboration of the Korea International Cooperation Agency (KOICA) and the Government of Korea. The main data centre is located at the Ministry of Home Affairs, Singha Durbar, Kathmandu.

A Disaster Recovery Site has been established in Hetauda to improve the security, backup, and availability of government and citizens' data in case of failures or disasters. The main purpose of GTDC is to integrate and securely store government data in a centralized system and support e-governance services in Nepal.

The objectives of GIDC are:

- Provide easy and reliable access to government IT services for the public
- Support job creation through the growth of information technology
- Develop a strong knowledge-based IT industry in Nepal
- Strengthen the implementation of e-Governance in the country
- Enable citizens to access government services electronically from anywhere

Electronic Transaction Act (ETA), 2063:

The Electronic Transaction Act 2063 is a law in Nepal that regulates electronic transactions. It deals with the use of computers, internet, and other electronic means to create, store, send, and verify digital information and records. In simple words, it makes electronic communication and online transactions legal, safe, and valid.

The objectives of ETA, 2063 are:

- To provide legal recognition for electronic records and digital signatures.
- To make electronic data generation, communication, and transmission reliable.
- To ensure secure and authentic electronic communication.
- To regulate and control computer and internet-based activities.

The scope of ETA, 2063 are:

- Creation and use of digital signatures.
- Control and punishment of cyber and computer-related crimes.
- Protection of intellectual property in electronic form.

- Regulation of electronic transactions and online activities.
- Establishment of regulatory bodies such as:
  - Office of Controller of Certification (OCC)
  - Certifying Authorities (CA)
  - Subscribers (users of digital certificates)

The major provisions of ETA, 2063 are:

- Legal recognition of electronic records and digital signatures.
- Rules for dispatch, receipt, and acknowledgment of electronic records.
- Regulation of certification authorities and their duties, rights, and functions.
- Government use of digital signatures and certificates.
- Provisions related to computer networks and internet service providers.
- Punishment for computer-related crimes (cyber-crimes).
- Establishment of an IT Tribunal for handling cases (both original and appeal jurisdiction).

Information Communication Technology (ICT) Policy, 2072:

Vision of ICT Policy 2072:
To transform Nepal into an information and knowledge-based society and economy.

Mission of ICT Policy 2072:
To promote the growth and development of the ICT sector as a key driver for sustainable development and poverty reduction in Nepal.

The goals / Objectives / Benefits of ICT Policy 2072 are:

- Improve Nepal's ICT development ranking at the international level.
- Achieve at least 75% digital literacy among citizens.
- Ensure broadband internet access for the majority of the population (about 90%).
- Expand internet access to the entire population.
- Increase ICT contribution to the national economy (around 7.5% of GDP).
- Provide 80% of government services online and improve government services using ICT.
- Promote e-procurement for transparency in public procurement.
- Develop G2G services for automated government systems (citizenship, land, etc.).
- Develop secure, reliable, and modern ICT infrastructure.
- Reduce digital divide and ensure universal access to ICT.
- Promote ICT education, skills, research, innovation, and digital literacy.
- Support startups, innovation, and ICT entrepreneurship including startup funding.
- Increase productivity in key economic sectors through ICT.
- Expand ICT applications in agriculture, health, tourism, and environment.
- Strengthen cybersecurity and data protection systems.
- Build Nepal into a knowledge-based digital society.

The strategies / Action Plan of ICT Policy 2072 are: **[IMP]**

- Development of ICT human resources.
- Integration of ICT in education and research.
- Development of ICT infrastructure and broadband access.
- Promotion of ICT industry and digital economy.
- ICT for e-governance and public service delivery.
- Expansion of ICT in agriculture, health, and tourism.
- Promotion of e-commerce and digital payments.

- Use of ICT in climate change and disaster management.
- Development of cloud computing and modern technologies.
- Ensuring cybersecurity and digital safety.

The areas of ICT Policy 2072 are:

- ICT in education, research, and skill development.
- ICT industry growth and investment promotion.
- ICT for government services and good governance.
- Development of telecommunications infrastructure.
- Expansion of rural and remote connectivity.
- Promotion of e-commerce and SMEs.
- ICT in agriculture, health, and tourism.
- Cybersecurity and safe digital environment.
- Promotion of gender equality in ICT use.
- Building confidence, standards, and interoperability in ICT systems.

Digital Signature:

A digital signature is a mathematical technique used to verify the authenticity and integrity of digital messages or documents. It ensures that the message is created by a verified sender (authentication) and that it has not been altered during transmission (integrity).

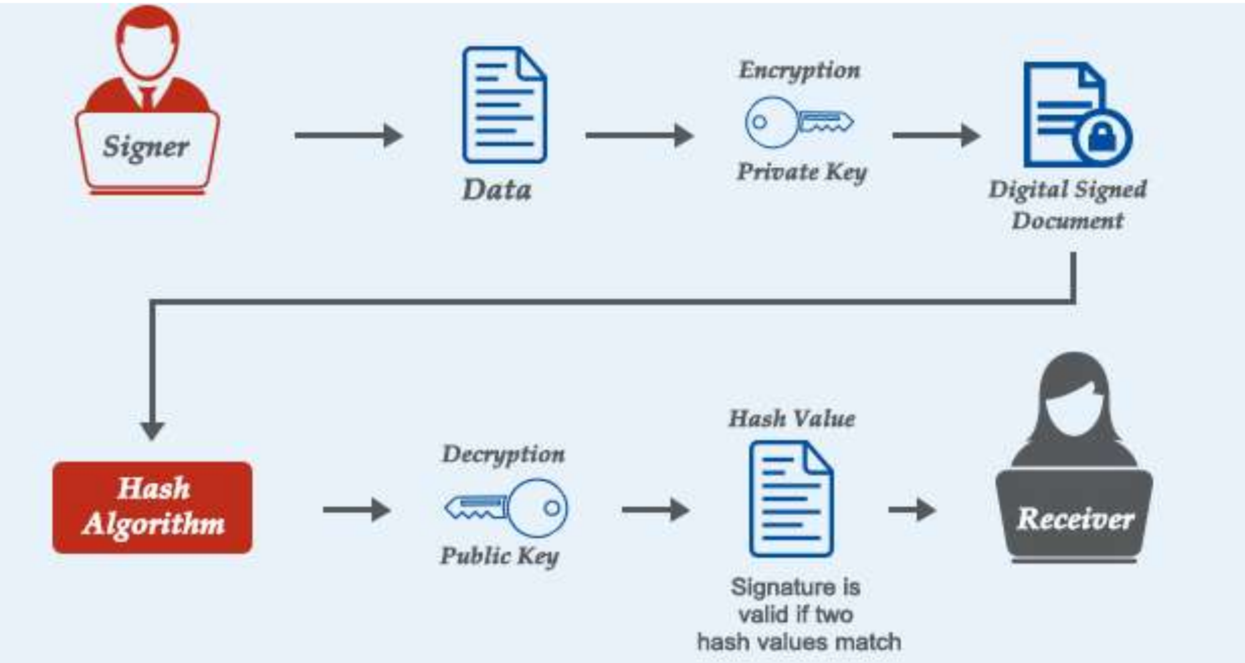


Fig: Working of Digital Signature

The working of a digital signature is based on Public Key Infrastructure (PKI), which uses a pair of keys: a private key and a public key. When a sender signs a document, a mathematical algorithm creates a hash value of the message and encrypts it using the sender's private key, forming the digital signature. This signature is attached to the document along with the time of signing. If any change is made to the document after signing, the signature becomes invalid. The receiver then uses the sender's public key to decrypt and verify the signature, ensuring the message is authentic and unchanged. A Certificate Authority (CA) is often involved to securely manage and verify the identity and keys used in the process.

The significance of Digital Signature:

- **Authenticity:** It verifies the identity of the sender or signer of the document.
- **Integrity:** It ensures that the document has not been changed or tampered with during transmission.
- **Non-repudiation:** It prevents the signer from denying their involvement in the signed document.
- **Security:** It enhances the security of electronic documents and transactions by reducing the risk of fraud and tampering.

The benefits of Digital Signature are:

- Digital signatures give legal recognition to electronic documents and contracts.
- It reduces costs by eliminating the need for paper, printing, and physical storage.
- It speeds up the approval and processing of documents and administrative work.
- It helps protect the environment by reducing paper usage.
- It allows secure and easy signing of contracts and agreements while verifying the identity of the signer.

The uses of Digital Signature are:

- **Document signing:**
  It verifies the identity of the sender and ensures the document is signed by the correct person.
- **Online services:**
  It is used in online services to verify user identity and ensure secure access.
- **E-procurement:**
  It is used in electronic procurement systems for submitting and verifying bids and official documents.
- **Data validation:**
  It helps in checking and verifying data before submission in online systems.
- **Certificate verification:**
  It confirms that the digital certificate is genuine and issued by a trusted authority.
- **Legal compliance:**
  It helps organizations follow legal rules, policies, and procedures for electronic transactions.
- **Authentication:**
  It ensures that the document or message truly belongs to the person who signed it.
- **Verification of originality:**
  It confirms that the document is original and has not been altered.

***

Unit 6

Recent Trends in E-Governances

Government 2.0: *[or, Gov 2.0]*

Government 2.0 (Gov 2.0) refers to the use of modern internet tools and collaborative technologies to make government more open, transparent, and efficient. It focuses on involving citizens, businesses, and government together to improve public services using digital platforms and open data.

Gov 2.0 means "putting government in the hands of citizens" by encouraging participation and interaction through online systems. It combines Web 2.0 technologies with e-government to allow people to access information, share ideas, and develop useful applications and services. In this system, the government mainly provides open data, web services, and digital platforms that support innovation and improve service delivery.

E-Democracy 2.0:

E-Democracy (Electronic Democracy), also called Digital Democracy or Internet Democracy, is the use of Information and Communication Technology (ICT) in political and government processes. It allows citizens to take part in decision-making, discussions, and policy-making through digital platforms such as the internet and mobile applications.

The term is believed to have been introduced by digital activist Steven Clift. E-Democracy uses modern ICT tools, including civic technology and government technology, to improve democratic participation. It enables all adult citizens to participate equally in proposing, developing, and shaping laws and public policies. In simple terms, it makes the democratic process more open, transparent, and citizen-focused using digital technology.

Open Data:

Open data refers to data that is freely available for everyone to access, use, and share without restrictions from copyright, patents, or other control systems. It promotes transparency and allows individuals, organizations, and governments to reuse data for different purposes such as research, innovation, and public services. The concept is closely related to other "open" movements such as open-source software, open government, open knowledge, and open science.

The principles of Open Data are:

- **Open by Default**: Data should be open and publicly available by default. It should only be restricted for valid reasons like privacy, security, or law.
- **Timely, Completed and updated**: Data should be released quickly and in full detail so it remains useful and up to date.
- **Easy Access & Usability**: Data should be easy to find, machine-readable, free of cost, and provided in open formats with clear licenses.
- **Standardized & Interoperable**: Data should follow common standards so different datasets can be easily combined and used together.
- **Transparency & Good Governance**: Open data improves transparency, helps citizens understand government work, and increases accountability.
- **Innovation & Inclusive Growth**: Open data supports innovation, business development, and solutions in areas like agriculture, climate, and public services.

The uses of Open Data are;

- Improves transparency.
- Encourages citizen participation in government activities.
- Empowers citizens by providing access to useful information.
- Helps in developing new products and services in the private sector.
- Promotes innovation and new ideas.
- Improves the efficiency of government services.
- Increases the effectiveness of public service delivery.
- Helps in evaluating the impact of policies and decisions.
- Supports discovery of new knowledge by combining and analyzing large data sets.

Mobile Governance: *[or, M-Governance]*

Mobile governance, or m-Governance, refers to the use of mobile devices and wireless technologies such as smartphones, laptops, and internet services to deliver government services. It is an extension of e-Government that helps improve communication and service delivery between the government, citizens, and businesses using mobile and wireless platforms.

The benefits of M-governance are:

- Reduces the cost of providing government services.
- Improves efficiency in government operations and service delivery.
- Helps modernize and transform public sector organizations.
- Provides greater convenience and flexibility for users.
- Improves access to government services for citizens.
- Reaches a large number of people through mobile devices.
- Encourages citizen participation and involvement in service delivery.
- Works effectively even without fixed or traditional infrastructure.

Open Standards for Web Presence:

Open standards for web presence are publicly available rules and guidelines that define how websites, applications, and digital systems should work and communicate with each other. These standards are developed through an open and collaborative process so that anyone can use, implement, and contribute to them without restrictions. They are important because they ensure that different systems, browsers, and platforms can work together smoothly (interoperability), reduce compatibility issues, and prevent dependency on a single company or technology. Open standards also promote fairness, transparency, innovation, and long-term sustainability of web technologies, making digital services more accessible and reliable for everyone.

Government Cloud Services (G-Cloud):

Government Cloud (G-Cloud) is a cloud computing platform that provides shared IT resources such as storage, computing power, and networking services to government organizations. It allows government bodies to use digital infrastructure without maintaining their own physical servers, making services more efficient, flexible, and cost-effective.

G-Cloud provides a standardized way to access and manage resources like virtual machines, networks, and storage. It also offers APIs and tools for easy and automated use of services. This helps improve efficiency and prevents users from depending on a single technology or vendor (avoids vendor lock-in).

Getting Access to G-Cloud:
To get access to G-Cloud, an organization must request an account from the Department of Information Technology (DoIT) or the concerned service provider by submitting a formal application letter. The following details are required:

- **Organization Name**: The official name of the organization that will use the G-Cloud services.
- **User Name**: The full name of the authorized person who will manage the account. This person acts as the administrator with full access.
- **Email ID**: A valid and official email address used for communication, notifications, and password recovery. It must be secure and active.
- **Resource Allocation**: The initial requirement of cloud resources such as CPU, RAM, storage, and IP addresses. Additional resources can be requested later as needed.
- **Organization Stamp**: The official stamp of the organization used for verification and approval of the request.

Open Source:

Open source refers to software or products whose source code is freely available for anyone to view, use, modify, and distribute. It allows users to access the design and development details of a product and make improvements or changes as needed. Open source is mainly used in software development under open-source licenses, but it can also apply to other types of digital content and collaborative work.

Open-source software is developed in a collaborative and decentralized way, where many developers contribute and review the code. Because of this community-based development, it is often more flexible, cost-effective, and long-lasting compared to software made by a single company. The open-source approach has also grown into a global movement that encourages sharing, innovation, and solving problems through community collaboration.

***

Unit 7

Case Study

Asian Development Bank (ADB):

ADB is an international financial institution established in 1996. It provides loans, grants, technical assistance, etc. Its main purposes are:

- Reduce poverty
- Support economic growth
- Improve infrastructure and technology
- Help developing countries like Nepal

ADB Website Link:
https://www.adb.org/

Extra:
ICT stands for Information and Communication Technology. It includes Internet, computers, Mobile communication, Networking systems, Digital Services, etc.

ICT Development Project by ADB in Nepal: *[Case Study]*

Introduction

The ICT Development Project in Nepal was supported by the Asian Development Bank (ADB) with the goal of improving the use of Information and Communication Technology (ICT) in government services, education, and rural communication. The project focused on developing e-governance, improving internet access, increasing digital literacy, and making government services more efficient and transparent.

Objectives of ICT Development Project:

The main objectives were:

- Develop ICT infrastructure in Nepal
- Promote e-governance services
- Improve access to information
- Connect rural areas with communication technologies
- Increase digital literacy
- Improve transparency and efficiency in government offices
- Encourage private sector participation in ICT

Major Areas of ICT Development Project:

The major areas of this project are:

- **E-Governance Development**
  This area focused on using digital systems in government offices instead of manual paperwork. It included online information systems, digital records, and electronic communication. As a result, government services became faster, more transparent, and easier for citizens to access.

- **Rural Telecommunications**
  The project worked to improve communication facilities in rural areas of Nepal. It expanded telephone and internet services and built communication infrastructure in remote regions. This helped people access education, health information, government services, and business opportunities.

- **Human Resource Development**
  ADB supported ICT education and training programs in Nepal. Government employees and students received training in computer skills, ICT management, and technical fields. The main aim was to create skilled manpower for Nepal's digital development.

- **Policy and Institutional Reform**
  The project also helped the government develop ICT-related policies and laws. It supported areas such as National ICT Policy, telecommunication regulation, cyber laws, and e-governance frameworks. These policies helped manage and promote ICT development in Nepal properly.

Role of ICT Development in E-Governance:
The ICT Development Project became an important step for e-governance in Nepal. The contribution to E-governance are:

- Digitalized government operations
- Improved communication among offices
- Helped citizens access information online
- Encouraged transparency and accountability

Advantages of ICT Development project:
The advantages of ICT Development project are:

- **Better Government Services:** Citizens received faster and more efficient government services.
- **Increased Transparency:** Digital systems helped reduce corruption and misuse of paperwork.
- **Improved Communication:** Communication between government offices became quicker and more effective.
- **Rural Development:** Rural and remote areas gained access to modern communication and information technologies.
- **Economic Growth:** ICT development created new business opportunities and increased employment.

Challenges faced:
The challenges in ICT Development project are:

- Many rural areas lacked proper electricity, internet, and communication facilities.
- Nepal faced a shortage of trained ICT professionals and technical manpower.
- Many people had limited knowledge and skills in using digital technology.
- Developing ICT infrastructure required a large amount of investment and funding.
- Nepal's mountainous geography made ICT expansion and connectivity difficult.

Outcomes / Achievements:
The project helped Nepal in the following ways:

- Expanded internet and telecommunication services across the country.
- Introduced digital government practices in public services.
- Improved ICT awareness among citizens and government staff.
- Developed and strengthened ICT infrastructure.
- Supported the implementation of e-governance initiatives.
- Created a strong foundation for future Digital Nepal programs.

National ID in Nepal: *[Case Study]*

The **National ID Card system in Nepal** is a government initiative designed to create a unique identity for every citizen using a digital database. It is implemented under the National ID and Civil Registration Act and managed by the Department of National ID and Civil Registration.

The **National ID (NID)** is a smart, biometric identity card that contains a citizen's personal and biological information. It is linked to a centralized national database and is used to uniquely identify each Nepali citizen.

The feature of National ID are:

- Contains a unique National ID number for each citizen
- Stores biometric data like fingerprints, facial image, and iris scan
- Includes personal details such as name, date of birth, address, and family information
- Linked to a central digital database system
- Designed as a smart card for digital verification

The objectives of National ID in Nepal are:

- To provide a single, unique identity to every citizen
- To prevent duplicate or fake identities
- To improve public service delivery
- To support e-governance and digital systems
- To make government records more accurate and secure

Working of National ID is given as:

- Citizens register their personal and biometric details at enrollment centers
- Data is verified and stored in a **central database system**
- A unique National ID number is generated
- The smart card is issued and can be used for identification and verification

The uses of National ID are:

- Accessing government services (passport, citizenship-related services, etc.)
- Banking and financial verification (KYC process)
- Social security and welfare programs
- Voting and national records management
- Digital authentication in e-governance systems

The benefits of National ID are:

- Reduces identity fraud and duplication
- Improves efficiency in government services
- Helps in better planning and policy-making using accurate data
- Strengthens security and digital governance
- Makes service delivery faster and more reliable

The challenges in National ID are:

- Data privacy and security concerns arise due to the storage of sensitive personal and biometric information.
- Limited technical infrastructure in rural areas makes enrollment and system access difficult.
- Low public awareness and digital literacy reduce effective participation in the system.
- The implementation of the National ID systems across the country has been slow, which delayed its full nationwide coverage and effectiveness.

Government Electronic Procurement System of Nepal (GEPS): *[Case Study]*

The Government Electronic Procurement System of Nepal (GEPS or e-GP system) is an online platform used by the Government of Nepal to manage public procurement processes digitally. It is implemented by the Public Procurement Monitoring Office (PPMO) to make government purchasing more transparent, efficient, and accountable.

GEPS is a web-based system where government offices can publish tenders, receive bids, evaluate proposals, and award contracts electronically instead of using traditional paper-based methods.

The objectives of GEPS are:

- To make the procurement process transparent and corruption-free
- To reduce paperwork and manual processes
- To increase efficiency and speed in public procurement
- To ensure fair competition among suppliers and contractors
- To improve accountability in government spending

The working of GEPS is:

- Government offices publish online tender notices
- Interested suppliers register and submit electronic bids (e-bidding)
- The system securely receives and stores bids
- Evaluation is done digitally based on set criteria
- Contracts are awarded and published online for transparency

The features of GEPS are:

- Online tender publishing and bidding
- Digital bid submission (e-bidding)
- Supplier registration system
- Electronic evaluation and contract awarding
- Centralized procurement database

The benefits of GEPS are:

- Reduces corruption and manual interference
- Saves time and reduces paperwork
- Increases transparency in government procurement
- Encourages wider participation from suppliers
- Improves efficiency and record management

The challenges of GEPS are:

- Limited technical knowledge among users
- Internet and infrastructure issues in remote areas
- System security and data management concerns
- Resistance to change from traditional methods

IT Park Kavre, Nepal: *[Case Study]*

The Information Technology Park (IT Park) Kavre, located in Kavrepalanchok District, is Nepal's first and only government-established IT Park. It was developed to promote the information technology sector, digital services, and ICT-based employment in Nepal.

It is situated about 23–30 km east of Kathmandu, with the aim of creating a dedicated hub for IT companies, startups, and digital innovation.

The main goal of the IT Park is to:

- Promote IT industry development in Nepal
- Attract IT companies and foreign investment
- Create employment opportunities for skilled youth
- Support software development and outsourcing services
- Develop Nepal as a regional IT hub

The key features of IT Park are:

- Large area of land with planned IT infrastructure
- Office buildings, business blocks, and residential facilities
- Designed for **software companies, data centers, and IT services**
- High-speed communication and networking facilities (planned)
- Government-supported ICT development zone

The importance of IT Park are:

- Helps decentralize IT activities outside Kathmandu Valley
- Supports the vision of **Digital Nepal**
- Can boost innovation, startups, and outsourcing industry
- Encourages skilled professionals to work inside Nepal

The challenges of IT Park are:

- Delayed operation and poor utilization
- Lack of strong policy support and incentives for companies
- Infrastructure and connectivity issues
- Limited attraction for private IT firms

current situation of IT park:
Although the IT Park was built with a large investment, it has not been fully operational for many years. It has remained largely underutilized, with only partial use of its facilities. Very few private IT companies have shown interest in establishing offices there. However, in recent years, efforts have been made to revive the park by involving universities, government agencies, and technology companies to make it more active and functional.

E-Village / Telecentre in Nepal:

An e-village (electronic village) or telecentre in Nepal is a rural ICT-based service center that provides internet access, computer facilities, and various digital services to people living in villages. It is mainly introduced to reduce the digital gap between urban and rural areas.

A telecentre is a public place where local people can use computers, the internet, printing, scanning, and other basic ICT services. It also helps people learn basic computer skills and access useful information related to education, health, and government services.

In Nepal, e-village or telecentre programs are supported by the government and different organizations to promote e-governance and improve services like education, communication, and health in remote areas. These centers are often connected through wireless or community networks.

The main purpose of telecentres is to bring digital services to rural communities, improve digital literacy, and support social and economic development through the use of ICT.

Smart City in Nepal:

The Smart City concept in Nepal is an urban development approach that uses ICT and modern technology to improve city services, infrastructure, and the quality of life of people. The main objective is to develop well-planned, efficient, and sustainable cities by using digital systems for better governance, infrastructure management, and service delivery. It also focuses on e-governance, reducing pollution, managing urban growth, and making cities more livable.

The concept uses ICT in areas like transportation, waste management, water supply, electricity, security, and public services. Online systems and digital platforms are used to make government services faster, easier, and more transparent. It also supports data-based planning and decision-making. In Nepal, cities like Budhanilkantha, Dhulikhel, Banepa, and Panauti are planned as smart cities, but none are fully developed yet and most are still in the planning or early stages.

Smart cities bring benefits like improved infrastructure, better public services, efficient transportation, and digital governance. However, challenges include lack of funding, weak ICT infrastructure, limited technical skills, slow implementation, and rapid urban growth.

Digital India Project:

The Digital India Project is a flagship program launched by the Government of India in 2015 to transform the country into a digitally empowered society and knowledge economy. The main objective of this project is to provide digital infrastructure as a basic utility to every citizen, deliver government services online, increase digital literacy, and promote transparency and efficiency in governance. It also aims to reduce corruption, improve connectivity in rural areas, and support the growth of a digital economy.

The project is based on three main components: creating digital infrastructure as a utility for every citizen, ensuring governance and services on demand, and promoting digital empowerment of citizens. These components help in improving internet access, delivering public services electronically, and building digital skills among people.

Several major initiatives have been introduced under this project, such as DigiLocker for storing digital documents, the UMANG app for accessing multiple government services, BHIM UPI for digital payments, and Common Service Centres (CSC) that provide digital access in rural areas. The project has also focused on expanding broadband and mobile connectivity across the country.

The Digital India Project has brought many benefits, such as faster and more convenient access to government services, reduced paperwork, improved transparency, better connectivity in rural areas, and the growth of the digital economy, startups, and employment opportunities. However, it also faces several challenges, including the digital divide between urban and rural areas, lack of internet infrastructure in remote regions, cybersecurity and data privacy issues, low digital literacy among some citizens, and the high cost of implementation.

***

Some Missing Topics

1. What are the key challenges in implementing E-governance in developing countries?

The key challenges in implementing E-governance in developing countries are:

- **Lack of ICT Infrastructure:** Many rural and remote areas do not have proper internet access, electricity, computers, or communication networks.
- **Low Digital Literacy:** Many citizens and even government employees may not have enough knowledge or skills to use digital technologies effectively.
- **Financial Constraints:** Developing countries often face limited budgets, making it difficult to invest in modern technology, maintenance, and training.
- **Resistance to Change:** Government staff and citizens may hesitate to shift from traditional paper-based systems to digital systems.
- **Cybersecurity and Privacy Issues:** Risks such as hacking, data theft, and misuse of personal information create challenges in maintaining secure systems.
- **Lack of Skilled Human Resources:** There may be a shortage of trained IT professionals and technical experts to manage e-governance systems.
- **Poor Policy and Legal Framework:** In some countries, laws related to cyber security, digital signatures, and electronic transactions are weak or not properly implemented.
- **Corruption and Political Instability:** Corruption, lack of transparency, and frequent political changes can delay or affect e-governance projects.
- **Language and Accessibility Barriers:** E-governance services may not be available in local languages, making them difficult for all citizens to use.
- **Lack of Public Awareness:** Many people may not know about available online government services and their benefits.

2. Describe the latest IT policy of Nepal in brief.

The Government of Nepal introduced the National Information and Communication Technology (ICT) Policy, 2072 to promote the development and use of information and communication technology in the country. The policy aims to build a knowledge-based society and support economic and social development through ICT.

The key features of latest IT policy of Nepal are:

- Expansion of broadband internet services in rural and urban areas.
- Promotion of digital literacy and ICT education.
- Development of IT parks and innovation centers.
- Encouragement of electronic transactions and online services.
- Use of open-source software and local language computing.
- Strengthening cyber laws and data security systems.
- Promotion of research and development in ICT.

3. What is Cyber Law? What are the various areas that cyber law covers?

Cyber Law is the branch of law that deals with the legal issues related to the use of computers, internet, digital communication, information systems, and electronic transactions. It provides rules and regulations to govern activities in cyberspace and helps prevent cyber crimes, protect digital data, and ensure safe and secure use of information technology.

The various areas that cyber law covers are:

- **Cyber Crimes:** Covers crimes such as hacking, phishing, online fraud, identity theft, cyber bullying, and spreading viruses.
- **Electronic Transactions:** Provides legal recognition to electronic records, digital signatures, and online transactions.
- **Data Protection and Privacy:** Protects personal and confidential information from unauthorized access and misuse.
- **Intellectual Property Rights (IPR):** Protects copyrights, trademarks, patents, and software from illegal copying or piracy.
- **E-Commerce Regulation:** Regulates online business activities, electronic payments, and consumer protection in digital markets.
- **Digital Signature and Authentication:** Ensures security and authenticity of electronic documents and communications.
- **Cyber Security:** Deals with protection of computer systems, networks, and data from cyber attacks and threats.
- **Internet Governance:** Regulates the proper and ethical use of the internet and online communication services.
- **Social Media and Online Content:** Controls illegal or harmful online content, including fake news, defamation, and offensive materials.
- **Domain Name and Website Issues:** Handles disputes related to domain names, website ownership, and cybersquatting.

4. Describe the need of cyber law.

Cyber Law is needed because most activities today are done using computers, internet, and digital systems, which create both opportunities and risks. It provides legal protection, security, and proper regulation for online activities. The need of cyber law:

- **To control cyber crimes:** Prevents and punishes crimes like hacking, phishing, and online fraud.
- **To give legal recognition:** Makes electronic records, digital signatures, and online transactions legally valid.
- **To protect data and privacy:** Secures personal and sensitive information from misuse.
- **To ensure safe online transactions:** Builds trust in e-commerce, online banking, and digital payments.
- **To regulate internet use:** Sets rules for proper and responsible use of the internet.
- **To protect intellectual property:** Prevents piracy and illegal copying of digital content.
- **To support e-governance:** Ensures secure and legal digital government services.
- **To maintain cyber security:** Protects systems and networks from cyber attacks.
- **To resolve online disputes:** Provides legal solutions for digital conflicts.
- **To build public trust:** Encourages safe and confident use of digital services.

5. Why is OLAP important in E-governance?

OLAP (Online Analytical Processing) is important in e-governance because it helps in fast analysis of large government data for better decision-making. OLAP is needed for:

- **Quick decision-making:** Helps government analyze data quickly for policies.

- **Data analysis:** Easily processes large and complex government data.
- **Better planning:** Supports planning of development programs and services.
- **Trend identification:** Helps find patterns like population, revenue, and service usage.
- **Improved efficiency:** Makes government work faster and more effective.

6. Explain the dissemination of Information technology.

Dissemination of Information Technology means the process of spreading or distributing ICT knowledge, tools, and services to people, organizations, and government so that everyone can access and use technology effectively. It helps in making digital services available to all areas, especially rural and remote regions, and promotes digital inclusion in society.

7. What is cyber-crime?

Cyber-crime refers to illegal activities that are carried out using computers, the internet, or other digital devices. It includes offenses such as hacking, online fraud, identity theft, phishing, spreading viruses, cyber bullying, and unauthorized access to data or systems. Cyber-crime can target individuals, organizations, or even government systems, causing financial loss, data breaches, and privacy violations. It is a serious issue in the digital world, and laws like cyber law are used to prevent and punish such activities.

8. What are the main challenges and courses that control e-governance in Nepal?

The main challenges and courses that control e-governance in Nepal are given as:

- **Weak ICT infrastructure:** Poor internet, electricity, and network facilities in rural areas.
- **Low digital literacy:** Many citizens and staff lack ICT skills.
- **Limited budget:** Insufficient funding for technology development and maintenance.
- **Resistance to change:** Preference for traditional paper-based systems.
- **Cyber security issues:** Risk of hacking, data theft, and system attacks.
- **Lack of coordination:** Poor communication between government offices.
- **Political instability:** Affects continuity of ICT projects and policies.

9. Which country has an efficient e-governance system and how?

A well-known example of an efficient e-governance system is **Estonia.**

Estonia uses a fully digital government system where almost all public services are available online. Citizens can access services like voting, tax filing, healthcare records, business registration, and banking through a secure digital ID system. The country uses strong cyber security, a paperless administration system, and fast internet connectivity nationwide. Because of this, government services are quick, transparent, and highly efficient, saving time and reducing corruption.

10. Define digital divide.

Digital divide refers to the gap between people who have access to modern information and communication technology (ICT) such as computers and the internet, and those who do not have access or cannot use them effectively due to lack of resources, skills, or infrastructure.

11. Write about E-governance in health and education.

E-governance in health and education means using information and communication technology (ICT) to improve the delivery, management, and accessibility of health and education services.

E-governance helps in managing hospital records, online appointment systems, telemedicine services, and digital health records. It allows patients to consult doctors online and access medical information easily. It also improves monitoring of diseases, vaccine programs, and public health data, making health services faster and more efficient.

E-governance is used in online admission systems, digital classrooms, e-learning platforms, and student record management. It helps students access study materials online and allows teachers to manage attendance, exams, and results digitally. It also supports distance learning and improves the quality and reach of education.

12. What are the changes in the citizen-governance relationship though e-governance.

Digital governance has changed the relationship between citizens and government by making it more direct, fast, and transparent. The changes are:

- **Direct access to services:** Citizens can now access government services online without visiting offices.
- **Increased transparency:** Online systems reduce corruption and make government activities more visible.
- **Faster service delivery:** Services like registration, tax payment, and applications are processed quickly.
- **Better communication:** Citizens can give feedback and complaints directly through digital platforms.
- **Citizen participation:** People can participate in decision-making through online surveys and systems.
- **Reduced middlemen:** Digital systems reduce dependency on agents or intermediaries.
- **Improved accountability:** Government work becomes more trackable and responsible.

13. What social impact does e-governance have on society?

E-governance has brought many positive social changes in society by using ICT to improve government services and communication.

- **Improved access to services:** People can easily access government services online from anywhere.
- **Increased transparency:** Reduces corruption and makes government work more open.
- **Time and cost saving:** Citizens do not need to travel or wait in long queues.
- **Better communication:** Improves interaction between government and citizens.
- **Digital inclusion:** Encourages people to use technology and become digitally aware.
- **Empowerment of citizens:** People can participate more in government decisions.
- **Improved quality of life:** Faster services in health, education, and administration improve daily life.

14. What is encryption? Write its types.

Encryption is the process of converting plain readable data (plaintext) into an unreadable format (ciphertext) using a secret key. It is used to protect data from unauthorized access during storage or transmission.

The types of Encryption are:

- **Symmetric Encryption:** In this type, the same key is used for both encryption and decryption. It is fast but less secure if the key is shared improperly.
- **Asymmetric Encryption:** In this type, two different keys are used: a public key for encryption and a private key for decryption. It is more secure but slower than symmetric encryption.

15. What is the utilization of electronic technology in the administration?

The utilization of electronic technology in administration is called E-Governance. E-governance means using computers, internet, and other ICT tools in government administration to deliver services, manage records, and communicate with citizens efficiently. It helps make government work faster, more transparent, and more effective.

16. What is the performance in NIC?

NIC (National Information Commission) is responsible for ensuring the right to information in Nepal.

The performance of NIC includes monitoring and implementing the Right to Information (RTI) Act, promoting transparency in government offices, and ensuring citizens can access public information easily. It also handles complaints related to denial of information and gives necessary orders to government bodies to provide information.

Overall, NIC plays an important role in improving accountability, transparency, and good governance in public administration.

17. How can e-governance security difficulties be withdrawn?

E-governance security problems can be reduced by applying strong technical and administrative measures:

- **Use of strong authentication:** Apply passwords, OTP, and biometric systems for secure access.
- **Encryption of data:** Protect sensitive information by converting it into unreadable form.
- **Regular system updates:** Update software to fix security bugs and vulnerabilities.
- **Use of firewalls and antivirus:** Protect systems from viruses, malware, and unauthorized access.
- **Cyber security laws:** Implement and enforce strong cyber laws and regulations.
- **User awareness training:** Educate employees and citizens about safe use of digital systems.
- **Backup and recovery system:** Keep regular backups to recover data after attacks or failure.
- **Access control system:** Limit system access only to authorized users.

18. What is the impact of e-learning sites on students?

E-learning sites have a major impact on students by changing the way they learn and access education.

- **Easy access to learning:** Students can study anytime and anywhere using the internet.
- **Flexible learning:** They can learn at their own speed and revise topics easily.

- **Better understanding:** Videos, animations, and interactive content improve learning.
- **Wide range of resources:** Students get access to global study materials and courses.
- **Cost-effective:** Reduces the need for extra books and tuition fees.
- **Self-learning habit:** Encourages independent study and self-discipline.
- **Distraction risk:** Sometimes students may get distracted by non-educational content online.
- **Dependence on internet:** Requires internet access, which may be a problem in remote areas.

19. Explain the relation between ICT and governance.

Information and Communication Technology (ICT) and governance are closely related because ICT helps improve the way government works and delivers services to citizens.

ICT supports governance by making government services faster, more transparent, and more efficient through digital systems. It helps in storing and managing data, communicating information quickly, and providing online public services. With ICT, governments can reduce paperwork, minimize corruption, and improve decision-making through better data analysis.

At the same time, good governance encourages the use of ICT to ensure accountability, transparency, and citizen participation. This combination is known as e-governance, where ICT is used to strengthen governance systems.

20. What is Open government data (OGD)?

Open Government Data (OGD) refers to government data and information that is freely available to the public in a usable and accessible format. It allows citizens, businesses, and organizations to access, use, and share government data without restrictions.

OGD promotes transparency, accountability, and innovation by enabling people to analyze government activities and create new services or solutions using public data. It also helps improve decision-making and public participation in governance.

21. Define data system infrastructure.

Data system infrastructure refers to the combination of hardware, software, networks, databases, and facilities used to collect, store, process, manage, and distribute data in an organized and secure way.

It provides the basic foundation for handling data in organizations and supports efficient information flow, decision-making, and digital services such as e-governance systems.

22. Explain issues in e-governance implementations and how they can be solved.

E-governance implementation faces several challenges that affect its effectiveness and success.

Issues:

- **Weak infrastructure:** Poor internet, electricity, and ICT facilities.
- **Low digital literacy:** Lack of skills among citizens and government staff.
- **High cost:** Expensive setup, maintenance, and upgrading of systems.
- **Security risks:** Threats like hacking, data theft, and cyber attacks.
- **Resistance to change:** People prefer traditional paper-based systems.
- **Lack of coordination:** Poor communication between government departments.
- **Legal and policy gaps:** Weak cyber laws and unclear regulations.

Solutions:

- **Improve infrastructure:** Expand reliable internet and electricity access.
- **ICT training:** Provide digital skills training to staff and citizens.
- **Increase investment:** Allocate proper budget for ICT development.
- **Strengthen security:** Use encryption, firewalls, and strong cyber laws.
- **Awareness programs:** Encourage people to adopt digital services.
- **System integration:** Connect all government departments digitally.
- **Policy reform:** Develop strong and updated ICT and cyber laws.

23. What are the pre-requisites of e-readiness for infrastructure development?

E-readiness means how prepared a country or organization is to use ICT for development and e-governance. For proper infrastructure development, some basic requirements must be fulfilled.

- **Reliable ICT Infrastructure:** Availability of internet, computers, telecommunication networks, and electricity is essential for digital services.
- **Skilled Human Resources:** Trained IT professionals and digitally literate users are needed to operate and manage systems effectively.
- **Financial Support:** Sufficient budget is required for building, maintaining, and upgrading ICT infrastructure.
- **Strong Policy and Legal Framework:** Clear laws and policies are needed to support ICT use, cyber security, and digital transactions.
- **Access to Technology:** Affordable and easy access to hardware, software, and internet services for all citizens.
- **Cyber Security System:** Protection of data and systems from hacking, misuse, and cyber threats.
- **Institutional Support:** Government institutions and organizations must coordinate and support ICT development.
- **Awareness and Acceptance:** Citizens should be aware and willing to use digital services.

24. Explain IT in judiciary and Electronic service delivery.

IT in the judiciary means the use of computers, internet, and digital systems in court activities and legal processes. It helps in case registration, digital record keeping, online hearings, and faster judgment processing. Courts can manage case files electronically, which reduces paperwork and delays. It also improves transparency and allows easy access to legal information for lawyers and citizens.

Electronic service delivery refers to providing government services to citizens through electronic means such as websites, mobile apps, and online portals. Services like tax payment, license renewal, citizenship application, and certificates can be accessed online. It reduces the need to visit government offices, saves time and cost, and improves efficiency and convenience.

25. Explain the revolutionary stages of e-governance that are perceived in any governmental organization.

E-governance develops gradually in different stages as government organizations adopt ICT to improve services and administration. The stages are:

1. **Web Presence Stage (Information Stage):**
   In this stage, government organizations create websites to provide basic information like notices, policies, and contact details. It is a one-way communication system.
2. **Interaction Stage:**
   Citizens can interact with government through email, online forms, and downloading documents. It allows limited two-way communication.
3. **Transaction Stage:**
   In this stage, citizens can complete services online such as paying taxes, applying for documents, and registering services. It enables full online transactions.
4. **Transformation Stage:**
   Government services become fully integrated and automated. Different departments are connected, providing seamless and efficient services through a single platform.
5. **Digital Governance Stage (Advanced Stage):**
   This is the most advanced stage where governance becomes fully digital, data-driven, transparent, and citizen-centric. Technologies like AI, big data, and mobile services are widely used.

26. If you are elected as an information technology Minister (IT minister), what are your master plans for implementing e-governance in Nepal? Explain in your own View.

If I were elected as an IT Minister, my main goal would be to make e-governance simple, accessible, and effective for all citizens. My master plan would focus on the following areas:

- **Strong ICT Infrastructure Development:**
  I would expand high-speed internet, electricity, and digital networks in all rural and urban areas to ensure equal access.
- **Digital Literacy for All:**
  I would launch nationwide ICT training programs for citizens, students, and government employees to improve digital skills.
- **One-Stop Government Portal:**
  I would develop a single integrated platform where all government services like tax, citizenship, licenses, and certificates are available online.
- **Cyber Security Strengthening:**
  I would implement strong cyber laws, data protection systems, and security frameworks to protect government and citizen data.
- **Paperless and Cashless Government:**
  I would promote fully digital offices by reducing paperwork and encouraging online payments and digital records.
- **Improved Inter-Department Coordination:**
  I would connect all government offices through a unified digital system for better data sharing and faster services.
- **Public Awareness Campaigns:**
  I would conduct awareness programs to encourage citizens to use digital services confidently.
- **Investment in ICT Sector:**
  I would attract both national and international investment to develop IT parks, innovation centers, and startups.

27. How can we use e-governance to provide efficient services to citizens? Explain with suitable examples.

E-governance uses ICT tools like computers, internet, and mobile systems to improve government service delivery, making it faster, easier, and more transparent for citizens. The ways to provide efficient services are:

- **Online Service Portals:**
  Governments provide services through websites where citizens can apply for documents, licenses, and certificates without visiting offices.
- **Digital Records Management:**
  All citizen data is stored electronically, which reduces paperwork and makes retrieval faster and more accurate.
- **Faster Processing of Services:**
  Automated systems reduce manual work, so services like tax filing or passport application are completed quickly.
- **Mobile and SMS Services:**
  Citizens can receive updates, notifications, and services through mobile phones.
- **Transparency and Tracking:**
  Citizens can track the status of their applications online, reducing corruption and delays.
- **Integrated Government Systems:**
  Different departments are connected, so data can be shared easily without repeated documentation.

Examples are:

- Online tax filing system where citizens can submit and pay taxes electronically.
- Online citizenship or passport application systems that reduce physical visits to offices.
- E-health systems where patients book appointments and access medical records online.
- E-learning platforms used in education for online classes and exams.

28. Give reason why IT structure is needed to implement e-governance. Explain network and computing infrastructure needed to implement e-governance.

IT structure is essential for e-governance because it provides the basic foundation for delivering digital government services. Without proper IT structure, online systems cannot function effectively.

- **Supports digital services:** It enables online delivery of government services like registration, tax payment, and applications.
- **Ensures data management:** Helps in storing, processing, and retrieving large amounts of government data efficiently.
- **Improves communication:** Connects different government offices and citizens through digital networks.
- **Increases efficiency:** Reduces manual work, paperwork, and delays in public services.
- **Ensures security:** Protects sensitive government and citizen data through secure systems.

Network Infrastructures for E-governance:
Network infrastructure refers to the communication system that connects government offices and users.

1. **Internet connectivity:** High-speed broadband and mobile internet for smooth online services.

2. **LAN (Local Area Network):** Connects computers within a government office.
3. **WAN (Wide Area Network):** Connects different government offices across districts and regions.
4. **Data communication systems:** Routers, switches, and communication lines for data transfer.
5. **Cloud network systems:** Used for centralized storage and easy access to government data.

Computing Infrastructures for E-governance:
Computing infrastructure refers to hardware and software systems required to run e-governance applications.

1. **Servers:** Store and manage government databases and applications.
2. **Computers and workstations:** Used by government employees to perform digital tasks.
3. **Databases:** Store citizen records, government data, and service information.
4. **Software systems:** Applications for online services, management, and processing.
5. **Data centers:** Secure facilities for storing and managing large-scale government data.

***